PDBSNet: Pixel-Shuffle Downsampling Blind-Spot Reconstruction Network for Hyperspectral Anomaly Detection

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***Abstract*— Recent years have witnessed significant advances of deep learning technology in hyperspectral anomaly detection (HAD). Among these methods, existing unsupervised learning techniques that exploit image reconstruction errors to detect anomalous targets not only generate backgrounds but also reconstruct anomalies to a certain extent, thus failing to recognize anomalies from the original hyperspectral image (HSI). In order to train an efficient background reconstruction network, this article proposes a new pixel-shuffle downsampling blind-spot reconstruction network (called PDBSNet) that operates in a self-supervised fashion. On the one hand, PDBSNet adopts a blind-spot architecture with the receptive field, whose center pixel is set as a blind spot (meaning that the network is unable to see the center pixel), and reconstructs the spectral information of the center pixel using its neighbors. Given that the spectral signatures of anomalous targets are significantly different from those of neighboring pixels, our network tends to generate high reconstruction errors for anomalous pixels and low reconstruction errors for background pixels, and the above characterization of blind-spot architecture is well-suited for the HAD task. On the other hand, we introduce the pixel-shuffle downsampling (PD) technique, which extends the application range of the blind-spot framework due to the fact that it breaks the spatial correlation between pixels in anomalous targets and improves the discrimination between the anomalous pixels and shuffled neighboring backgrounds. Taken together, our PDBSNet integrates the blind-spot architecture and the PD strategy, effectively enhancing the feature representation of the background and weakening the feature expression of anomalies. It is expected to increase the reconstruction error of anomalous objects, while the differences between the original HSI and the reconstructed one are taken as anomaly scores. Extensive experiments on four datasets reveal that PDBSNet is competitive and remarkable concerning other state-of-the-art detectors.**

***Index Terms*— Deep learning (DL), hyperspectral image (HSI), image reconstruction, neural network, self-supervised learning (self-SL).**

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I. Introduction

HYPERSPECTRAL imaging technology records the fine spectral characteristics of ground objects [1], [2], [3], [4], and subtle spectral differences can be used to identify ground materials [5], [6], [7]. As one of the most popular tasks, hyperspectral anomaly detection (HAD) aims to identify anomalous targets in hyperspectral images (HSIs) without the aid of any prior knowledge about targets [8], [9].

Targets occupying relatively small areas with spectral features that deviate from those of most surrounding objects and are typically considered anomalies [10], [11]. Making full use of the acquired data to design a reasonable background model is essential for locating anomalous targets. However, due to limitations in imaging conditions, the distinguishability between background and anomalies is reduced, making it challenging to model backgrounds precisely and to separate anomalies from them. Researchers have proposed many HAD approaches to address core issues related to the recognition of abnormal attributes, which can be divided into two categories, namely, traditional machine learning-based and deep learning (DL)-based methods.

A. Traditional Machine Learning-Based Methods

The Reed–Xiaoli (RX) anomaly detector is the first statistical-based HAD method and assumes that the background satisfies a multivariate Gaussian distribution [12]. The background information of the HSI is also employed to estimate the probability density function. Meanwhile, the outliers are detected by calculating and comparing the Mahalanobis distance between each pixel and the background distribution. The detectors that model the background using all pixels and the neighborhoods of the testing pixel are tagged as the global RX (GRX) and the local RX (LRX) [13], respectively. Later,

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to boost the ability of the HAD method to recognize anomalous targets in complicated scenes, some improved derivatives have been developed from different sides. For example, the segmented RX (SRX) [14] requires K-Means clustering on the full HSI and uses the neighboring pixels of each pixel for background estimation before performing HAD. Then, Tao et al. [15] transformed all pixels in the original spectral domain into the fractional Fourier domain (FrFD) to enhance the difference between background and anomalies, further improving the detection accuracy. Besides, Liu et al. [16] proposed the 2S-GLRT, which adopts a sliding dual-window strategy to determine whether the central pixel is anomalous and responds to the presence of multipixel anomalies.

Afterward, representation-based methods, such as collaborative representation (CR), sparse representation (SR) [17], [18], [19], [20], [21], and low-rank representation (LRR) [22], [23], [24], [25], [26], gradually gained widespread attention. The CR-based detector (CRD) [27] dissects the case where each pixel is linearly represented by its outer window pixels. Then, it measures the reconstruction residuals to serve as the probability of anomaly for the central pixel. Since anomalous pixels easily contaminate the background dictionary constructed by CRD, Hou et al. [28] designed a background purification model to remove outliers from the dictionary, effectively alleviating the impact of potential anomalies. When viewed from the perspective of LRR, an HSI can be compartmentalized into low-rank and sparse components [29], corresponding to backgrounds and anomalies, respectively. Zhang et al. [30] adopted the Mahalanobis distance between the testing pixel and the low-rank matrix to represent the abnormal values of pixels. Meanwhile, Ma et al. [31] transferred the original HSI to the FrFD and combined it with low-rank and sparse matrix decomposition techniques. In addition to the matrix decomposition-based methods described above, LRR can also utilize the original HSI to construct a background dictionary [32]. For this purpose, the residual difference between the original HSI and the reconstructed one can be employed to assess anomalies. Meanwhile, alternative dictionary construction approaches have emerged, e.g., clustering algorithms based on K-Means [33], [34] or Mean-Shift [35], among others.

B. DL-Based Methods

DL has steered a course in the analysis of HSIs with its brilliant capacity to mine and interpret higher order information contained in complicated scenes [36], [37]. There are two broad family methods for HAD based on DL: similarity measurement- and reconstruction-based models. Similarity measurement-based approaches require an accurately labeled (reference) classification HSI to establish tens of thousands of training sample pairs, which satisfies the needs involved in the tuning of network parameters. Li et al. [38] trained a metric network with paired samples, which can output the difference value between each pixel and the outer window pixels in the test HSI. Then, Rao et al. [39] proposed a new Siamese network as the backbone and measured the similarity scores of the testing pixels and the surrounding ones at the hidden layer level. Nevertheless, the detection performance of these kinds of networks highly depends on the quantity and quality of training data. The annotation process requires technical expertise, which limits the flexibility of similarity measurement-based models from a practical viewpoint. In contrast, reconstruction models based on unsupervised learning (USL) do not need annotated data and employ the entire HSI as the training data. Such methods typically rely on the autoencoder (AE) or generative adversarial networks (GANs) to reveal deep intrinsic spectral properties in the HSI. Bati et al. [40] introduced the AE model into HAD, and this milestone work is based on the following core idea: since anomalous pixels account for a small proportion of the HSI, most of which are background pixels, the model does not yet fit anomalies adequately. This leads to the phenomenon that the reconstruction effect of background pixels is better than that of anomalous pixels. Naturally, the reconstruction error can represent the degree of anomaly of the pixel to be measured. To mine the spatial-contextual information in HSIs, Fan et al. [41] embedded graph regularization into the hidden layer of AE (RGAE) and redesigned the loss function with consideration of anomalies and noise. Furthermore, a fully convolutional AE-based automatic HAD (Auto-AD) method was devised by Wang et al. [42], achieving promising performance in an end-to-end training and inference manner. To facilitate the AE to generate background distributions more exhaustively and further reconstruct the background adequately, the discriminator is added to the AE (constituting GAN) by applying a Gaussian distribution [43] or a coarse screening of background samples before training the network is required [44], thereby increasing the separability of backgrounds and anomalies. Wang et al. [45] chose the parameter-free Mahalanobis distance to obtain relatively pure background samples and further integrated spatial and spectral information through a two-stream network.

C. Motivation and Contribution

As mentioned earlier, similarity measurement-based methods suffer from difficulties such as training sample pair construction and model generalization, which are somewhat incompatible with the unsupervised HAD task. Meanwhile, reconstruction-based models with the input as the target are committed to extracting low-level features between background and anomalies [46], [47], and a well-trained network can also partially reconstruct anomalies [48]. Once the anomalous targets are ready to be reconstructed, the reconstruction error of anomalies will be reduced to be in keeping with that of backgrounds. Hence, the ability of the model to extrude anomalies from the original HSI is relatively limited. In a nutshell, existing detectors are more or less limited by the number of samples and how well the model fits the anomalies, thus affecting the detection accuracy.

To this end, this study creatively introduces the self-supervised learning (self-SL) paradigm into the HAD field. Unlike USL-based approaches, the self-SL technique requires additional considerations: first, the input data and corresponding supervision information are elaborated without labeled samples; and second, the pretext task is

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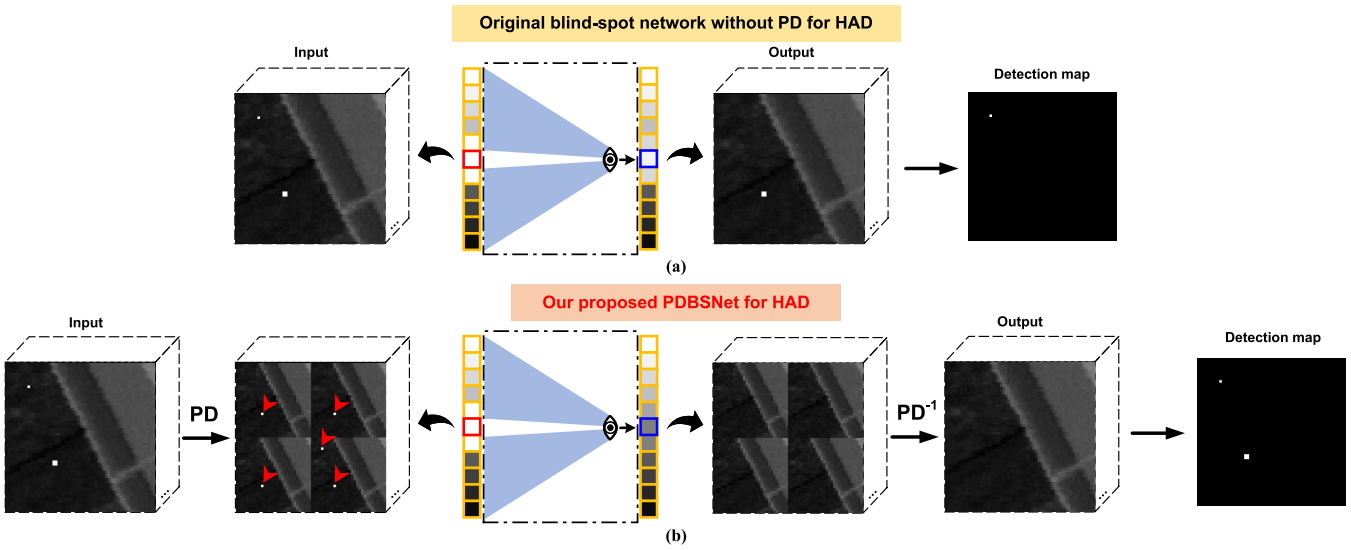

Fig. 1. HAD methods based on the blind-spot scheme predict the spectral information of each pixel using only the content of its neighbors (blue areas seen by the eye) in a self-supervised manner. Those bright pixels (anomalies) that differ significantly from the surrounding pixels will fail to be reconstructed, and the reconstructed error is employed as the final detection map. (a) Blind-spot network is directly adopted for HAD, and it is very likely that anomalous targets with certain spatial sizes will be reconstructed, resulting in unsatisfactory detection results. (b) Our newly proposed PDBSNet (introduced the PD strategy) extends the application scope of the blind-spot architecture since it breaks the spatial dependency between anomaly pixels (indicated by the red arrow), thereby highlighting the significant differences between these pixels and shuffled backgrounds.

designed according to the target task requirements, aiming at mining more purposeful features from unlabeled data and hopefully getting the best of both areas [49]. So far, self-SL methods have been studied in several fields of HSI interpretation, to name a few: denoising [50], unmixing [51], super-resolution [52], and classification [53], with satisfactory results. As one of the most well-known self-SL techniques, the architecture of blind-spot networks suits the HAD task quite well [54]. Without seeing the pixel information (blind spot) in the center of the receptive field, the blind-spot network series [55], [56] predict the intensity of blind spots using those surrounding pixels. If the center is an outlier with notable differences from its neighbors, it has little chance of being restored. Therefore, we are motivated to exploit a self-SL-based blind-spot mechanism for HAD. Specifically, in order to set blind spots in the input HSI, an architecture based on the masked convolution and dilated convolution (MDC) module is designed, called the dilated blind-spot network (DBSN). At this point, this network takes the original HSI as the label and then outputs the prediction data for each blind spot by completely extracting the spatial–spectral features of its neighbors. Due to the fact that the spectral characteristics of anomalies are significantly different from those of neighboring pixels, the reconstruction error of anomaly pixels is greater than that of background pixels. Hence, the trained network weakens the feature representation for anomalies and can be regarded as a superb background reconstruction model. However, the spectral differences between pixels inside an anomalous target with a certain spatial size are small, and the model might be able to predict details about anomalies whose reconstruction error will lead to a decrease, thus affecting the final detection accuracy, as shown in the top of Fig. 1. Then, a pixel-shuffle downsampling (PD) strategy [57], [58] is suggested to lessen the spatial correlation between anomalous pixels inside the target and enhance the distinguishability between the anomalies and the surrounding backgrounds, which will further broaden the application range of the proposed blind-spot network (see the bottom of Fig. 1). Finally, the degree of anomaly of the pixels in the HSI can be assessed using the reconstruction error of the pixels. Compared with currently existing HAD methods, the contributions of our proposed PD blind-spot reconstruction network (PDBSNet) are threefold.

1) To improve the model's comprehension of limited unlabeled samples, a new blind-spot mechanism-based framework, PDBSNet, is proposed for HAD purposes in a self-SL manner. This study will shed some valuable light on the problem of insufficient training samples in the HAD field.
2) To enhance the capability of the model to express and reconstruct background features, the DBSN model based on a blind-spot scheme is designed, whose character of using surrounding pixels to predict the blind-spot information is very suitable for the HAD task. Specifically, the proposed reconstruction model fully employs the joint spatial–spectral characteristics of the contextual pixels to predict the intensity of each pixel without seeing the spectral information of corresponding input pixel. Naturally, since background objects are predominant in HSIs and there is a significant difference between the anomalies and the surrounding backgrounds, anomalous samples exhibit a larger reconstruction error than background samples, which is more favorable for recognizing anomalies from the original HSI.
3) To extend the application range of the serial blind-spot network for HSIs with targets of slightly larger size, the PD strategy is introduced. This can break the spatial

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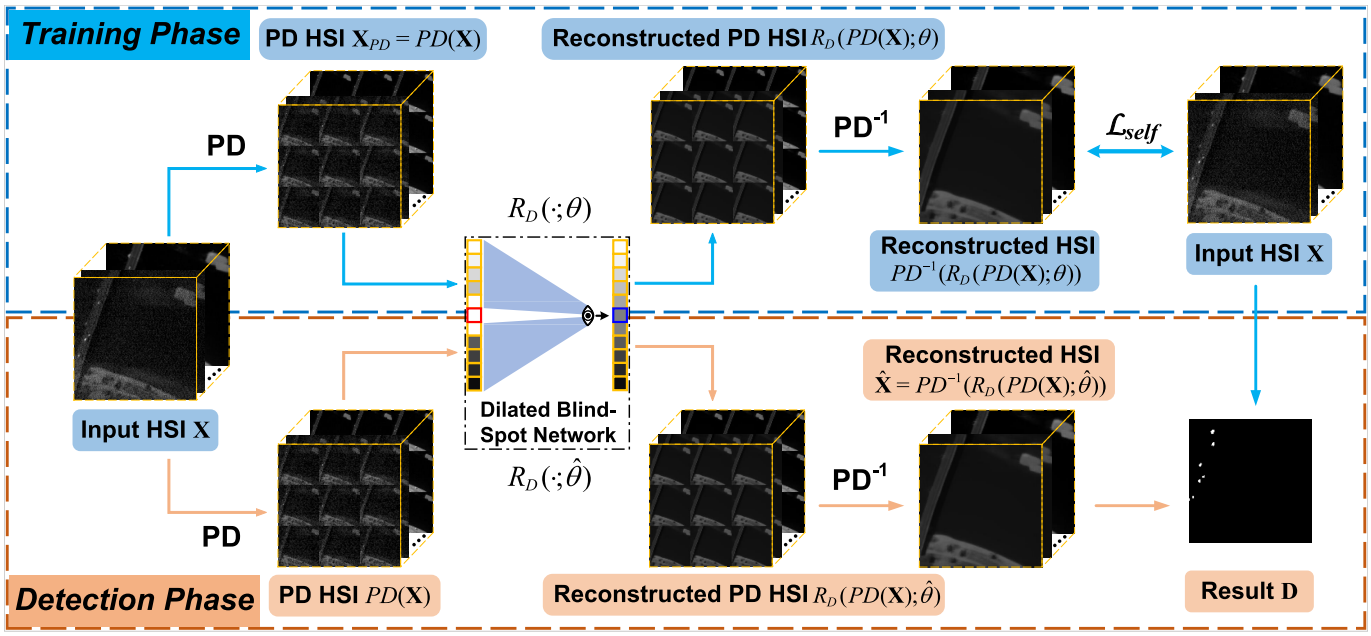

Fig. 2. Flowchart of the proposed PDBSNet method for HAD.

dependency between the pixels inside the anomalous target and improve the discriminability and separability of anomalies from the shuffled neighboring backgrounds, effectively eliminating the opportunity for the self-supervised model to reconstruct anomalous objects.

The remainder of this article is structured as follows. In Section II, the details of the proposed PDBSNet are described. Extensive experimental results and comparisons between PDBSNet and state-of-the-art methods are presented and evaluated in Section III. Finally, the conclusion is given in Section IV.

II. Proposed Method: PDBSNet

A. Overview

We propose a new self-SL-based approach to detect anomalies. Its overall framework is shown in Fig. 2. More specifically, our PDBSNet approach can be divided into two stages.

1) *Training Phase:* First, a DBSN model based on MDC modules is built in a blind-spot fashion. Then, subsampling in the PD process is performed on the original HSI, and the downsampled HSI is input into the DBSN. Finally, the PD-inverse operation is implemented on the output of the network to reconstruct the HSI, and the objective function of PDBSNet is set to the reconstruction loss based on the $L1$ norm.
2) *Detection Phase:* This phase exhibits the same behavior as the data processing during the training phase, where the original HSI is fed into the trained PDBSNet model to capture the expected background HSI, and the reconstruction error is employed as the result of HAD. These contents will be described in detail next.

B. Problem Formulation

To begin with, we denote an HSI as $\mathbf{X} \in \mathbb{R}^{R\times C\times L}$, where R, C, and L are the row number, column number, and spectral dimension (here, the number of channels) of the HSI, respectively. We aim to train a background image reconstruction network in a self-supervised scheme, i.e., requiring no background-label information and only using the observed image $\mathbf{X}$. Our network learns the feature representation of the background, yielding low reconstruction errors for background pixels and high reconstruction errors for anomalous targets. Consequently, the reconstruction errors can indicate abnormal levels of pixels and help identify anomalous targets.

C. Training Phase

This section mainly introduces the PDBSNet and its training phase, which integrates the PD and self-SL-based DBSN model proposed in this article.

1) Pixel-Shuffle Downsampling: The PD operation has been used to reduce the spatial correlation between adjacent pixels in an image. To further increase the difference between the anomalous pixels and the surrounding backgrounds, a more appropriate input is generated for the DBSN model. The top of Fig. 3 demonstrates the diagram of PD with a stride factor of 2. The given HSI is decomposed into four subimages, which are tiled together and recorded here as PD$(\cdot)$. Meanwhile, the PD-inverse operation (i.e., pixel-shuffle upsampling) is shown at the bottom of Fig. 3 and is denoted as PD$^{-1}(\cdot)$.

2) Dilated Blind-Spot Network: First, we reiterate that the core concept of blind-spot networks is that the value of each pixel in the output image needs to erase the influence of the input value at the same position. To satisfy this condition, we set up a DBSN model based on the MDC module, and its detailed structure is shown in Fig. 4. In the following, we will discuss the principle of this model.

The basic component of the DBSN model is the MDC module that contains a masked convolution and a dilated convolution. Fig. 5 shows the simple MDC module, in which we ignore the channel dimension. Moreover, since paddings and

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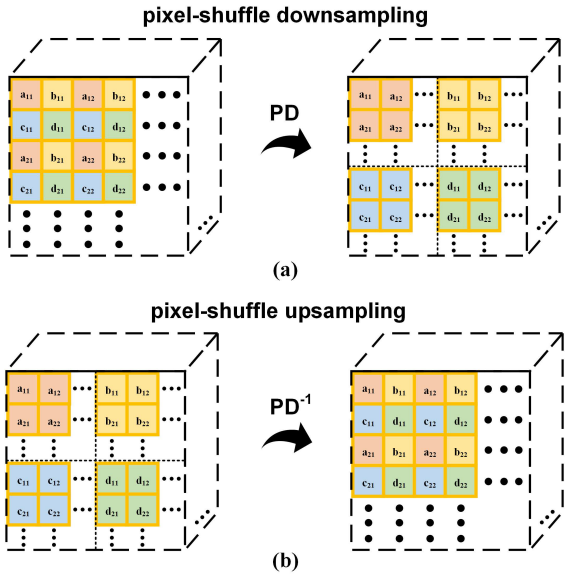

Fig. 3. Diagram of (a) PD$(\cdot)$ (PD) and (b) PD$^{-1}(\cdot)$ (pixel-shuffle upsampling) with a stride factor of 2.

stride ones are set, the output feature maps of each layer have the same spatial size. First, $\mathbf{X}$ and $\mathbf{w}$ are adopted to represent the input image and 3×3 convolutional kernel, respectively. The first layer is the centrally masked convolution, which introduces a 3×3 binary mask $\mathbf{m}$ with a center element of 0 and the rest of 1. The centrally masked convolution is expressed as

$$\mathbf{f}^1 = \mathbf{X} * (\mathbf{w} \odot \mathbf{m}) \tag{1}$$

where $\mathbf{f}^1$ represents the feature map of the first layer and symbols $*$ and $\odot$ are the convolution operator and multiplication by elements, respectively. The second layer is a dilated convolution with dilation factor $d = 2$. In a similar vein, the blind-spot constraint condition can also be matched by stacking dilated convolutional layers with $d = 3$ on the 5×5 centrally masked convolution. In the following, we explain why MDC-like structures qualify for the blind-spot mechanism.

In Fig. 5, taking the 9×9 input $\mathbf{X}$ as an example, each pixel is denoted as $\mathbf{X}_{i,j}$, $i \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ and $j \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$, where $\mathbf{X}_{0,0}$ is the pixel value of the center position, described the blind spot. First, the feature map $\mathbf{f}^1$ is obtained by operating the centrally masked convolution on $\mathbf{X}$ using (1). Clearly, values, including $\mathbf{f}^1_{0,0}$ and $\mathbf{f}^1_{i,j}$, $i \in \{-4, -3, -2, 2, 3, 4\}$ and $j \in \{-4, -3, -2, 2, 3, 4\}$, are not relevant to $\mathbf{X}_{0,0}$. However, $\mathbf{f}^1_{i,j}$, $i \in \{-1, 1\}$ and $j \in \{-1, 1\}$, are calculated with $\mathbf{X}_{0,0}$. The blind-spot mechanism requires that $\mathbf{f}^1_{i,j}$, $i \in \{-1, 1\}$ and $j \in \{-1, 1\}$, are avoided when calculating the central pixel value of the feature map $\mathbf{f}^2$, which prompts us to employ dilated convolution with dilation factor $d = 2$. When we obtain the feature map $\mathbf{f}^2$, it can be found that its central pixel value $\mathbf{f}^2_{0,0}$ is obtained by $\mathbf{f}^1_{i,j}$, $i \in \{-2, 0, 2\}$ and $j \in \{-2, 0, 2\}$ (dark blue squares in $\mathbf{f}^1$). As mentioned earlier, these values are not impacted by $\mathbf{X}_{0,0}$ but are related to the corresponding receptive field in $\mathbf{X}$ (blue squares in red rectangle). Furthermore, we find that $\mathbf{f}^2_{i,j}$, $i \in \{-2, 0, 2\}$ and $j \in \{-2, 0, 2\}$ are also irrelevant to $\mathbf{X}_{0,0}$. Consequently, the dilated convolution can continue to be adopted in the subsequent layers.

In fact, skip connection and 1×1 convolution will not destroy the blind-spot constraint either. Hence, we can create the DBSN model by leveraging centrally masked convolutional, dilated convolutional, and 1×1 convolutional layers. As shown in Fig. 4, DBSN begins with a 1×1 convolutional layer followed by two MDC module branches. The upper (lower) branch consists of a 3×3 (5×5) centrally masked convolutional layer, three 1×1 convolutional layers, and six dilated convolution modules with $d = 2$ ($d = 3$). In a dilated convolution module, a rectified linear unit (ReLU) function and a 1×1 convolutional layer are stacked upon a dilated convolutional layer. Meanwhile, the input and output feature maps are also summed at the pixel level (the same as below) to achieve skip connections. Then, the feature maps from the two MDC branches are concatenated. Finally, three 1×1 convolutional layers are applied at the end of DBSN to produce the output HSI. We denote the proposed DBSN model as R_D, parameterized by θ.

3) Integration PD and DBSN: The original HSI $\mathbf{X}$ is first decomposed into factor_train^2 sub-HSIs, assuming that the PD operation in the training phase has a stride factor factor_train. Next, the shuffled HSI $\mathbf{X}_{PD} \in \mathbb{R}^{R \times C \times L}$ obtained after PD is a tiling of these sub-HSIs $\mathbf{X}_{\text{sub}} \in \mathbb{R}^{R/\text{factor_train} \times C/\text{factor_train} \times L}$. Note that if $\mathbf{X}$ contains anomalous targets of a certain size, then selecting an appropriate factor_train will cause the anomalous pixels in the targets to be scattered in each subimage of $\mathbf{X}_{\text{PD}}$. This diminishes the spatial correlation between anomalous pixels while increasing the discriminability and separability between anomalies and neighboring pixels. The above process is expressed as follows:

$$\mathbf{X}_{\text{PD}} = \text{PD}(\mathbf{X}). \tag{2}$$

We reiterate that the DBSN model treats each pixel in the input HSI $\mathbf{X}_{\text{PD}}$ as a blind spot and only utilizes the spatial–spectral characteristics of neighboring pixels to reconstruct the spectral information of blind spots. As anomalies occupy a low proportion in HSI and most targets develop into isolated points after the PD operation, their neighborhoods are relatively homogeneous background areas, and the background objects do not contain any anomalous properties. Suppose that the network regards anomaly outliers as blind spots, and there is a high probability that an estimated value that satisfies the background distribution will be produced, thus eliminating the anomalous targets and making the PDBSNet evolve into a background reconstruction network.

The shuffled image, PD$(\mathbf{X})$, is fed into our background reconstruction network, $R_D(\cdot; \theta)$, followed by a PD-inverse operation (i.e., pixel-shuffle upsampling), and then, we obtain a reconstructed image, denoted as PD$^{-1}(R_D(\text{PD}(\mathbf{X}); \theta))$.

The parameters of our reconstruction network are trained using the following ℓ_1 loss function:

$$\begin{aligned} \hat{\theta} &= \arg\min_{\theta} \mathcal{L}_{\text{self}} \\ &= \arg\min_{\theta} \left\| \text{PD}^{-1}(R_D(\text{PD}(\mathbf{X}); \theta)) - \mathbf{X} \right\|_1. \end{aligned} \tag{3}$$

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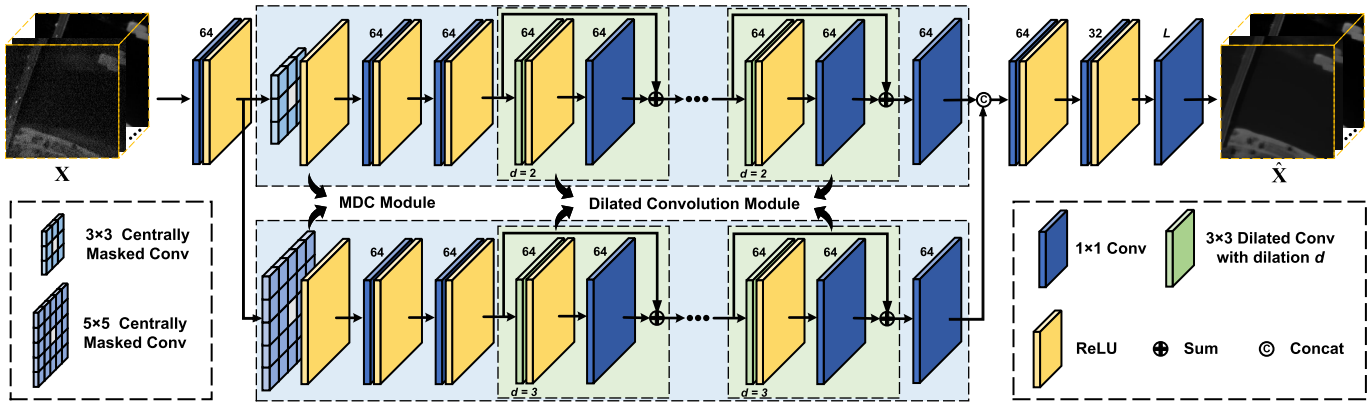

Fig. 4. Details of the proposed DBSN structure (the numbers above Conv, such as 64, represent the number of output channels).

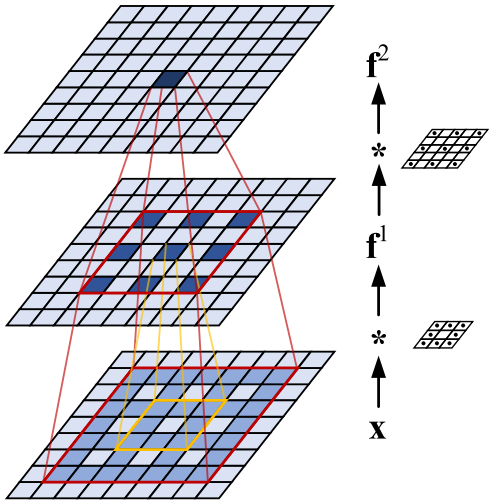

Fig. 5. Simple MDC module based on blind-spot scheme. The red rectangles indicate the receptive field area of the feature map $\mathbf{f}^1$ and the input $\mathbf{X}$ corresponding to the center of the feature map $\mathbf{f}^2$, and the yellow rectangle denotes the receptive field area of the input $\mathbf{X}$ corresponding to the center of the feature map $\mathbf{f}^1$. Meanwhile, the light squares in the receptive field areas of $\mathbf{f}^1$ and $\mathbf{X}$ represent the blind spots, and the remaining blue squares of different depths need to be calculated together to get the central pixel value of $\mathbf{f}^2$. Naturally, the actual receptive field of $\mathbf{X}$ corresponding to the center pixel of $\mathbf{f}^2$ does not contain information about the corresponding center position in $\mathbf{X}$, which fits the blind-spot mechanism, i.e., each pixel in the output HSI is reconstructed using only the surrounding pixels corresponding to the input pixel, and the network does not see the central pixel value of the receptive field.

Algorithm 1 Proposed PDBSNet for HAD

Input: original HSI $\mathbf{X} \in \mathbb{R}^{R\times C\times L}$, parameters: 1) training stride factor $factor_train$ of PD; 2) test stride factor $factor_test$ of PD.
Output: Final detection result $\mathbf{D} \in \mathbb{R}^{R\times C}$.

Stage1: Training Phase;
Obtain the shuffled HSI $\mathbf{X}_{PD} \in \mathbb{R}^{R\times C\times L}$ using the PD operation with a stride factor $factor_train$ by (Eq. 2);
Acquire the output $R_D(PD(\mathbf{X});\theta)$ of the DBSN model R_D, parameterized by θ, and R_D takes $\mathbf{X}_{PD}$ as input;
Restore $R_D(PD(\mathbf{X});\theta)$ to the full-sized reconstructed image $PD^{-1}(R_D(PD(\mathbf{X});\theta))$ using the PD-inverse operation;
Train a reconstruction network R_D by minimizing (Eq. 3);
Gain the optimal network R_D, parameterized by $\hat{\theta}$;
Stage2: Detection phase;
Obtain the shuffled HSI $PD(\mathbf{X})$ using the PD operation with a stride factor $factor_test$;
Acquire the output HSI $R_D(PD(\mathbf{X});\hat{\theta})$ by feeding $PD(\mathbf{X})$ into the well-trained R_D;
Restore the reconstructed HSI $\hat{\mathbf{X}} \in \mathbb{R}^{R\times C\times L}$ using the PD-inverse operation by (Eq. 4);
Calculate the degree of anomaly $D_{i,j}$ of each pixel in $\mathbf{X}$ by (Eq. 5);

D. Detection Phase

In the inference phase, analogously, we perform the PD operation with a stride factor factor_test in $\mathbf{X}$ and then fed the shuffled image into the well-trained network, parameterized by $\hat{\theta}$. After implementing the PD-inverse operation, we obtain a reconstructed image as follows:

$$\hat{\mathbf{X}} = \mathrm{PD}^{-1}(R_D(\mathrm{PD}(\mathbf{X});\hat{\theta})). \tag{4}$$

After the above blind-spot mechanism-based training process in a self-SL manner, the proposed PDBSNet becomes a credible background reconstruction network such that the anomalous pixels in the reconstructed image also approximate the surrounding background without restoring the original shape faithfully. Accordingly, the reconstructed HSI can be considered as the background HSI, and anomalies have significant reconstruction errors relative to the background. Finally, the anomaly detection result is achieved using (5) based on the reconstruction error

$$D_{i,j} = \left\|\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\right\|_2 \tag{5}$$

where $\mathbf{x}_{i,j} \in \mathbb{R}^{L\times 1}$ and $\hat{\mathbf{x}}_{i,j} \in \mathbb{R}^{L\times 1}$ stand for the pixels at location (i, j) of $\mathbf{X}$ and $\hat{\mathbf{X}}$, respectively, and $D_{i,j}$ denotes the anomaly probability values of the corresponding location pixels, which constitutes the final detection map $\mathbf{D} \in \mathbb{R}^{R\times C}$. The main steps of the proposed PDBSNet are summarized in Algorithm 1.

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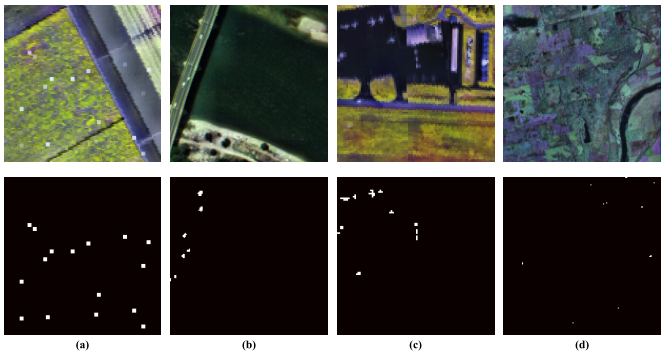

Fig. 6. Pseudo-color images and ground truth maps of the four considered HSI datasets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

TABLE I
DETAILS OF THE EXPERIMENTAL DATASETS

Data set	Location	Sensor	Image size	Resolution
HSI-I	Salinas	AVIRIS	$120 \times 120 \times 204$	3.7m
HSI-II	Pavia	ROSIS-03	$150 \times 150 \times 102$	1.3m
HSI-III	Gainesville	AVIRIS	$100 \times 100 \times 191$	3.5m
HSI-IV	Indiana	Hyperion	$150 \times 150 \times 155$	10m

III. EXPERIMENTAL RESULTS

In this section, we conducted several experiments to corroborate the superiority of the proposed PDBSNet qualitatively and quantitatively.

A. Datasets

The first dataset, HSI-I, is a synthetic image acquired from the Airborne Visible Infrared Imaging Spectrometer (AVIRIS). We referred to the anomalous target implantation approach following the linear mixture model [59], [60], [61] in [62], which is written as

$$\mathbf{n} = f_a\mathbf{a} + (1 - f_a)\mathbf{b} \tag{6}$$

where $\mathbf{a}, \mathbf{b}, \mathbf{n} \in \mathbb{R}^{L\times 1}$ denote the spectral vectors of the anomaly, background, and final mixture of the two, respectively, and f_a represents the abundance fraction of $\mathbf{a}$. The original background image and anomaly spectrum (building) of HSI-I were selected from the Salinas scene, and 16 anomaly objects are randomly embedded into the subimage, each with a spatial size of 3×3. Note that the corresponding abundance f_a for these targets starts at 0.05 and increases by 0.04 accordingly.

The remaining three datasets are all real scenes. Among them, the first real image (HSI-II) was derived from the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor over a beach area of Pavia, where the anomalies mainly consist of some vehicles. The second scene, marked as HSI-III, was captured by the AVIRIS, which covered an urban area in Gainesville. The last scene (HSI-IV) was captured by the Earth Observing One (EO-1) Hyperion sensor, and the anomalous targets were chiefly storage bins and roofs [63], [64]. More detailed characteristics of the above four images are listed in Table I. In addition, Fig. 6 shows these datasets' pseudo-color images and ground-truth maps of anomalous targets.

B. Comparison Algorithms and Evaluation Metrics

1) Comparison Algorithms: The ten well-established detectors selected in the experiment embrace five approaches based on local dual window (i.e., LRX [13], SRX_Local [14], CRD [27], CRDBPSW [28],[1] and 2S-GLRT [16][2]) and five global-based approaches (i.e., GRX [12], FrFE [15],[3] FEBPAD [31],[4] RGAE [41],[5] and Auto-AD [42][6]). In the above methods, based on the state-of-the-art DL architectures, RGAE and Auto-AD are designed to strengthen the reconstructing effect of the model on the background, and both adopt residual spectral errors between the original and reconstructed images to measure the abnormal level of pixels. This is consistent with the motivation of our proposed PDBSNet. Ergo, the differences between the detection results of these three approaches need to be observed to emphasize the effectiveness of the self-SL-based model.

2) Evaluation Metrics: We employed well-known criteria to assess the detection performance of different methods, i.e., statistical separability map, receiver operating characteristic (ROC), and the resulting area under the ROC curve (AUC) [65]. The ROC curve gives the relationship between the probability of detection and false alarm, revealing the detection effect of the method holistically. The corresponding AUC score can quantitatively display the accuracy indicator. An excellent detector takes along an ROC curve in close proximity to the top-left corner, with an AUC close near 1. In addition, the separability boxplot depicts the statistical range of the anomalous response values at the pixel locations of the background and anomaly categories. It distributes them in two boxes (blue for the background and red for the anomaly), each with a statistical range of 10%–90%. Naturally, if the algorithm has a predominant ability to suppress backgrounds and separate anomalies, the blue box will be at the bottom and narrow in width, with a large gap from the red box.

C. Detection Performance for Different Methods

The HAD colored maps of the 11 considered detectors on four datasets are visualized in Figs. 7–10, where the first one is the ground-truth of anomalies, and the rest are the detection results for different methods. In addition, Figs. 11 and 12 show the ROC curves and separability boxplots of all methods whose AUC values are recorded in Table II. Note that the optimal results were marked in bold letters (the same as below). Next, the performance of these traditional and DL-based detectors was tested and compared from multiple levels.

As it can be seen in Figs. 7–10, for the synthetic and real scenarios datasets, the proposed PDBSNet has relatively ideal results in the tradeoff of background suppressibility

[1]https://github.com/zephyrhours/Hyperspectral-Anomaly-Detection-CRDBPSW

[2]https://github.com/zephyrhours/Hyperspectral-Anomaly-Detection-2S-GLRT

[3]https://github.com/xudongzhao461/Hyperspectral-Anomaly-Detection-by-Fractional-Fourier-Entropy

[4]https://github.com/FGH00292/Hyperspectral-anomaly-detection-via-FEBPAD

[5]https://github.com/FGH00292/Hyperspectral-anomaly-detection-with-RGAE

[6]https://github.com/RSIDEA-WHU2020/Auto-AD

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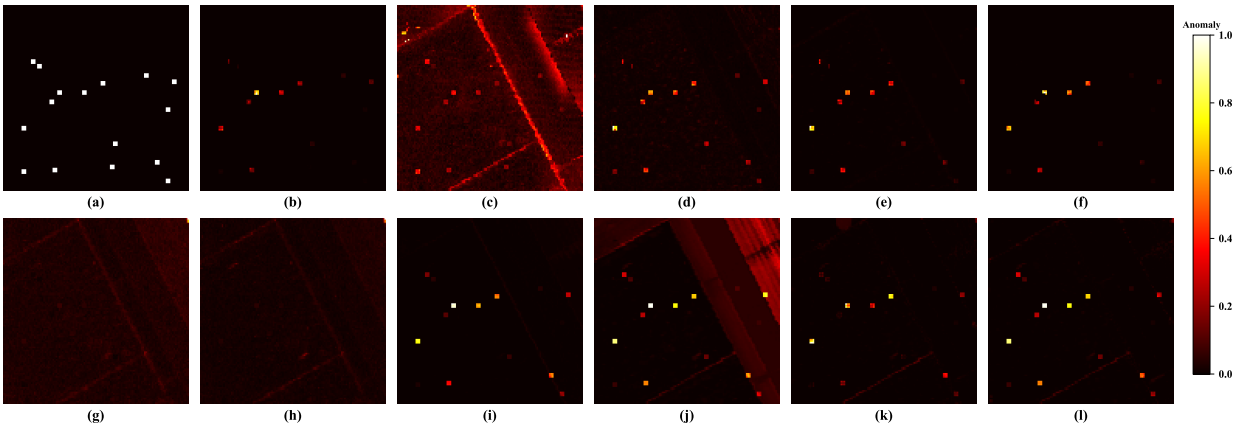

Fig. 7. HAD colored maps obtained by different algorithms on the HSI-I dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) FEBPAD. (j) RGAE. (k) Auto-AD. (l) PDBSNet.

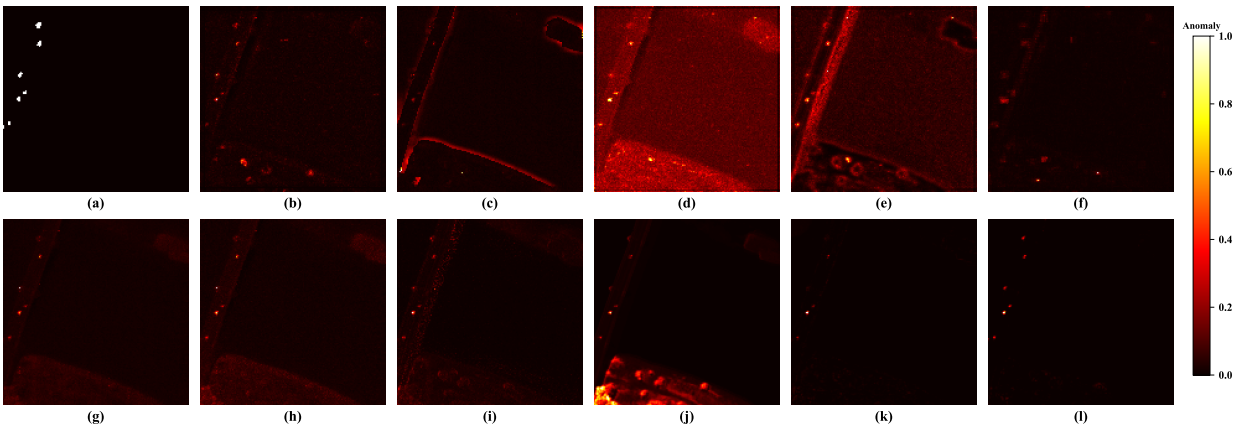

Fig. 8. HAD colored maps obtained by different algorithms on the HSI-II dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) FEBPAD. (j) RGAE. (k) Auto-AD. (l) PDBSNet.

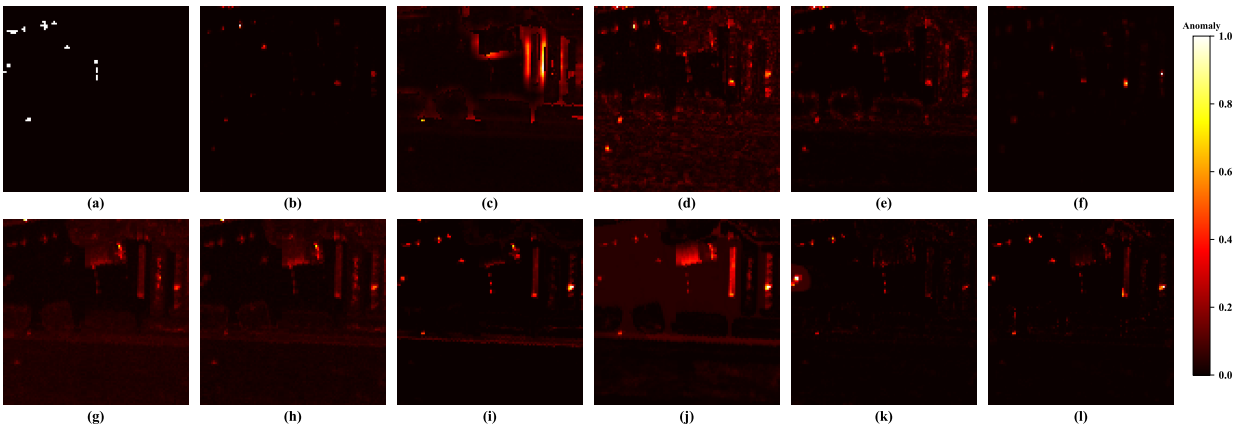

Fig. 9. HAD colored maps obtained by different algorithms on the HSI-III dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) FEBPAD. (j) RGAE. (k) Auto-AD. (l) PDBSNet.

and anomaly detectability, and its detection results are closest to the ground truth. Considering the HSI-I dataset as an example, GRX and FrFE do not work well because anomalous targets are hardly visible in the result maps, but background objects possess high anomaly probability values. Although SRX_Local and RGAE can detect some targets, they also retain some background structural information, thus generating more false alarms. The remaining detectors can adequately restrain the background, which reduces the risk of misjudging the background, yet our PDBSNet has higher abnormal response values to embedded pixels than other methods, hinting that PDBSNet is more likely to separate the targets from the background. Another example is the HSI-II dataset. Although CRD, CRDBPSW, and RGAE can highlight anomaly targets, they also outline the boundaries of background objects, and the background suppression is

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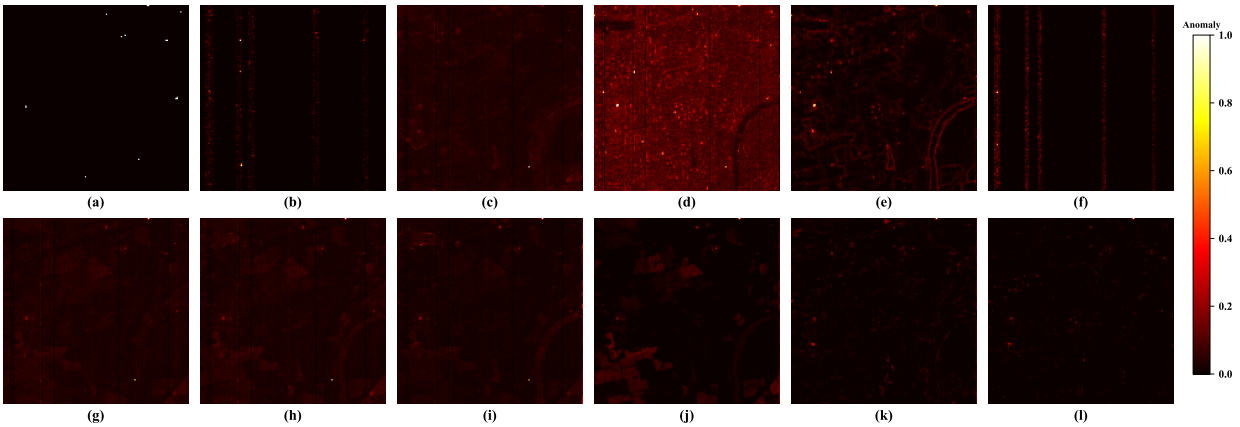

Fig. 10. HAD colored maps obtained by different algorithms on the HSI-IV dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) FEBPAD. (j) RGAE. (k) Auto-AD. (l) PDBSNet.

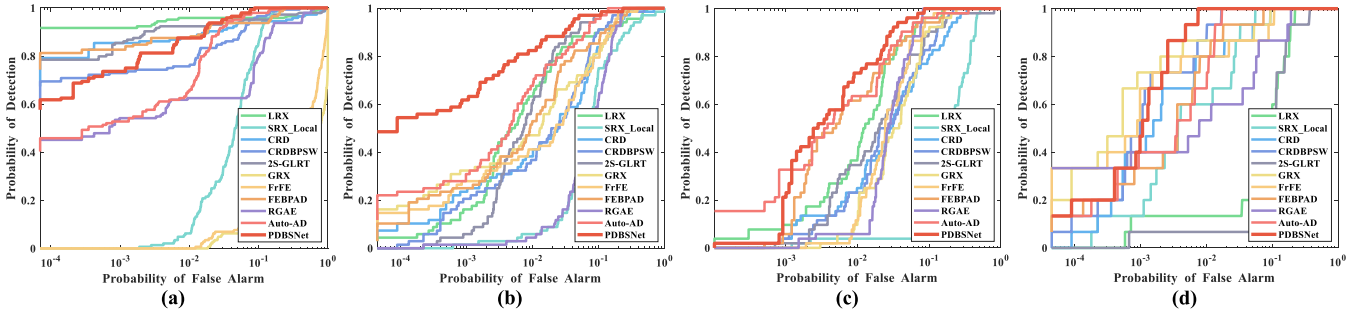

Fig. 11. ROC curves for different detectors on the four considered datasets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

TABLE II
AUC VALUES OF THE 11 CONSIDERED DETECTORS ON DIFFERENT DATASETS

Data Set	Detectors and AUC values										
	LRX	SRX_Local	CRD	CRDBPSW	2S-GLRT	GRX	FrFE	FEBPAD	RGAE	Auto-AD	PDBSNet
HSI-I	0.9747	0.9458	0.9642	0.9846	0.9779	0.3798	0.5270	0.9874	0.9365	0.9887	**0.9950**
HSI-II	0.9590	0.8795	0.9407	0.9552	0.9802	0.9538	0.9498	0.9637	0.9094	0.9820	**0.9900**
HSI-III	0.9767	0.7614	0.9393	0.9522	0.9489	0.9513	0.9645	0.9794	0.9665	0.9787	**0.9892**
HSI-IV	0.9018	0.9870	0.9877	0.9968	0.8676	0.9896	0.9850	0.9908	0.9602	0.9943	**0.9983**
Average	0.9531	0.8934	0.9580	0.9722	0.9437	0.8186	0.8566	0.9803	0.9432	0.9859	**0.9931**

unsatisfactory. SRX_Local is challenging to locate anomalies and is easily disturbed by the background. LRX, 2S-GLRT, GRX, FrFE, and FEBPAD can identify most of the targets, but it has the possibility to mistake backgrounds for anomalies. Auto-AD and PDBSNet can reduce false alarms to a large extent. Nevertheless, Auto-AD does not perform as prominently as PDBSNet in enhancing anomalous characteristics. Moreover, similar situations are seen in the results from the other two datasets.

Indeed, in addition to the visual effect of the HAD colored map, other metrics are needed to qualitatively and quantitatively evaluate the performance of different algorithms. Fig. 11 shows the ROC results of these four considered datasets. In most cases, the red curve of PDBSNet is above other curves and closest to the top-left corner, which outperforms other methods as a whole. Not surprisingly, as shown in Table II, the AUC scores of PDBSNet beat those of the ten comparison detectors. Fig. 12 further shows the strong separability ability of PDBSNet. Although the anomaly boxes of PDBSNet are not always in a high position for different datasets, their background boxes are narrow enough to be within a small range, implying that the model can swimmingly restrain the background. The two boxes of PDBSNet exist at a small degree of intersection on the HSI-II dataset, but the boxes of the rest methods intersect more severely. For the other three datasets, there is a large gap between the two boxes of PDBSNet. Evidently, this is consistent with the aforementioned visual analysis results, indicating that our model is capable of extracting anomalies from different HSIs.

Given all of that, from the HAD result maps, our proposed PDBSNet is a showcase for relatively superior visual effects. Besides, the analysis of the ROC curves, AUC values, and separability boxplots manifests that PDBSNet obtains the highest average AUC of 0.9931, which is 0.0072 higher than

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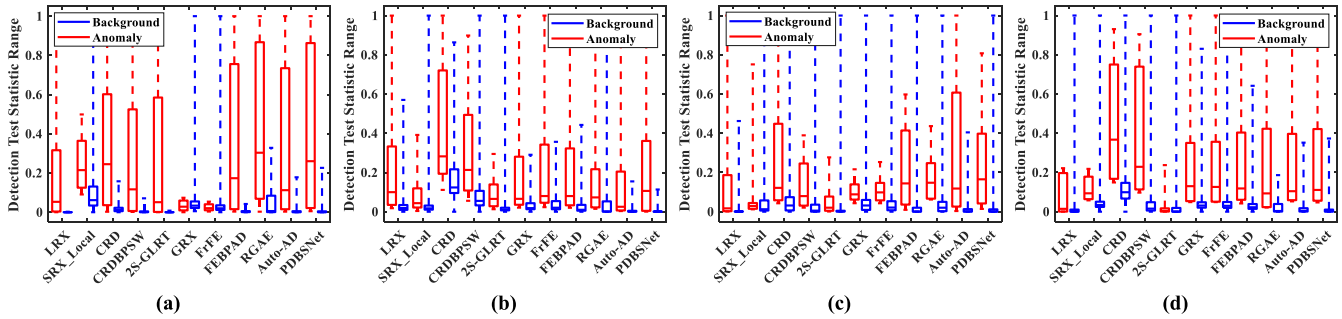

Fig. 12. Separability maps for different detectors on the four considered datasets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

the average AUC of the second-place Auto-AD, as shown in Table II. Meanwhile, the false alarm rate of PDBSNet is typically retained at a low level, further guaranteeing that the equilibrium between background suppressibility and anomaly detectability exists.

D. Parameter Analysis

The proposed PDBSNet involves two parameters to be analyzed: the stride factor factor_train and factor_test of PD during the training and inference phases. It is worth noting that the feasible values of factor_train and factor_test are consistent and need to be evenly divisible by the spatial size of HSI. Among them, the strides for the corresponding four datasets are set to {2, 3, 4, 5}, {2, 3, 5, 6}, {2, 4, 5}, and {2, 3, 5, 6}. The AUC values under different parameter settings are shown in Fig. 13. Next, the effects of factor_train and factor_test on the model detection performance are analyzed.

From Fig. 13, it is found that the performance of PDBSNet when factor_train and factor_test are the same is superior to that when the two are different since matching data distribution between test and training samples plays a key role. The appropriate stride factor needs to be chosen based on the spatial size of the anomalous targets. The four HSIs in the experiment contained targets of different sizes. According to the experimental results, we fixed the stride factor to 3, 3, 5, and 2 (factor_train is equal to factor_test).

E. Ablation Study

This section analyzes the effectiveness of the adopted self-SL and PD strategies. We take the traditional USL-based reconstruction model as the baseline, which adopts the same framework as the proposed DBSN model in this article, except that the masked convolution and the dilated convolution are replaced with standard 2-D convolutional kernels. The proposed PDBSNet first innovatively introduced the self-SL technique, which is embodied in the design of a DBSN model based on the blind-spot scheme. It is here described as PDBSNet without PD. Meanwhile, the PD strategy is also introduced to break the spatial correlation between pixels in the anomalous target, further extending the applicability of DBSN and strengthening the detection performance. The AUC values of these three models are compared on four datasets in Table III. It can be seen that the self-SL-based PDBSNet without PD has certain advantages over the USL-based approach.

TABLE III
AUC VALUES OF ABLATION STUDY ON DIFFERENT DATASETS

Data Set	Detectors and AUC values		
	USL-based	PDBSNet without PD	PDBSNet
HSI-I	0.9835	0.9852	**0.9950**
HSI-II	0.9780	0.9840	**0.9900**
HSI-III	0.9823	0.9849	**0.9892**
HSI-IV	0.9886	0.9920	**0.9983**

However, the performance improvement is relatively limited. This is because the datasets contain large-sized anomalous targets, and the spectral differences between pixels inside the target are small, so there is a high probability that the blind-spot network (PDBSNet without PD), which reconstructs each pixel value by using its neighboring pixels only, will reconstruct most of the anomalous pixels (especially the center anomaly pixels), thereby affecting the detection accuracy. This inspires us to adopt the PD strategy to improve the discriminability of anomalies from the shuffled neighboring backgrounds. Therefore, PDBSNet with PD can more significantly boost the ability of the self-SL-based blind-spot model to recognize anomalies from HSI.

F. Performance Analysis Between Target Size and Stride Factor

To verify the relationship between the optimal stride factor of PD and the target size, we embedded 2×2 anomalous targets with an abundance of 0.3 on the background subimage of HSI-I at regular intervals, and the target size was set to 1×1 to 9×9. At the same time, the stride factor (factor_train is equal to factor_test) was set to {1, 2, 3, 4, 5, 6, 8}. Next, the AUC scores obtained by the proposed PDBSNet on the nine synthetic datasets under different stride factors are listed in Table IV. We declare in advance that, when the stride is set to 1, PDBSNet degenerates to a self-SL-based model that does not adopt the PD strategy (same as PDBSNet without PD in Section III-E).

We found that the model without the PD strategy (with a stride factor of 1) exhibited varying degrees of performance degradation on HSIs with a target size greater than 4×4, and the larger the target size, the lower the corresponding AUC value. Meanwhile, when the stride is small (e.g., 2 and 3),

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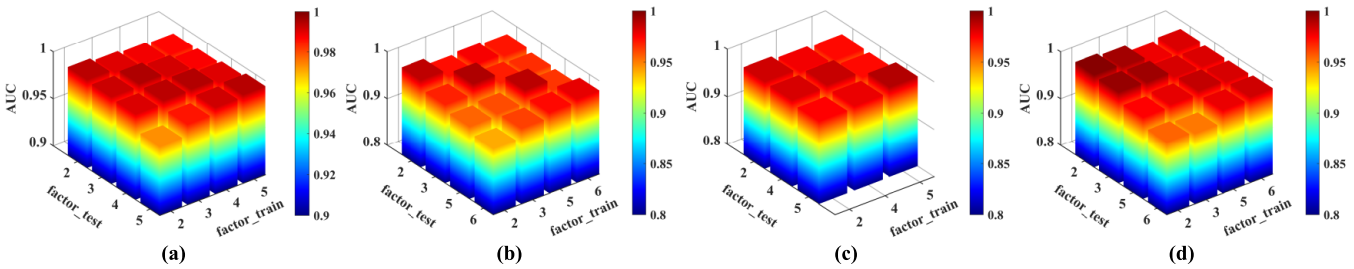

Fig. 13. AUC values of the proposed PDBSNet with different stride factors (factor_train, factor_test) on the four considered datasets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

TABLE IV
AUC VALUES OF THE PROPOSED PDBSNET WITH DIFFERENT PARAMETERS ON THE SYNTHETIC DATASET

Stride factor	Target size and AUC values								
	1×1	2×2	3×3	4×4	5×5	6×6	7×7	8×8	9×9
1 (PDBSNet without PD)	1.0000	0.9999	0.9998	0.9996	0.9896	0.9898	0.9874	0.9749	0.9698
2	0.9999	0.9999	0.9999	0.9996	0.9906	0.9757	0.9692	0.9533	0.9448
3	1.0000	0.9999	1.0000	0.9999	0.9998	0.9990	0.9894	0.9536	0.9340
4	1.0000	0.9999	1.0000	1.0000	0.9999	0.9998	0.9899	0.9619	0.9464
5	1.0000	1.0000	1.0000	0.9999	1.0000	0.9999	0.9997	0.9961	0.9782
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9987	0.9897
8	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9999

TABLE V
RUNNING TIMES (IN SECONDS) OF THE 11 CONSIDERED DETECTORS

Data Set	Detectors and running times										
	LRX	SRX_Local	CRD	CRDBPSW	2S-GLRT	GRX	FrFE	FEBPAD	RGAE	Auto-AD	PDBSNet
HSI-I	37.4267	17.4677	2.3189	5.3003	37.8787	0.0844	15.2690	3.8415	0.0462	**0.0289**	0.0419
HSI-II	17.4160	2.9611	2.9411	26.7408	22.7698	0.0486	11.7432	2.9318	0.0307	**0.0299**	0.0469
HSI-III	20.9789	2.8974	2.2356	5.2544	22.7819	0.0493	9.5050	2.4713	**0.0207**	0.0209	0.027
HSI-IV	36.4680	9.9309	3.0655	14.6764	34.808	0.0891	23.8665	5.4840	0.0412	**0.0329**	0.0529

a similar behavior occurs with PDBSNet on HSIs with a larger target size. However, as the stride of PD increases, the applicability of PDBSNet becomes wider, and in particular, PDBSNet can always achieve the optimal performance when the target size and the stride factor are close. This is because the PD strategy reduces the dependency between anomalous pixels and increases the separability of anomalies and shuffled surrounding backgrounds, thus making full use of the self-SL-based blind-spot framework for HAD. Therefore, we suggest selecting the stride factor of PD according to the spatial size of anomalous targets for accurate detection results.

G. Comparison of Inference Times

The inference time of different detectors on four considered datasets is reported in Table V to assess the computational complexity of each model. Except for Auto-AD and PDBSNet, implemented on a running environment with Python 3.9.12, Pytorch 1.12.1, and CUDA 11.3 by using GeForce RTX 2080 Ti GPU, the remaining methods were performed using MATLAB 2019b on an Intel[7] Core[8] i5-11400F CPU @ 2.60 GHz and 32 GB of RAM. Compared with traditional methods that employ some tricks, such as sliding dual window and fractional Fourier transform, GRX gains the shortest computation time with a concise design principle, but its detection performance is unsatisfying. The three DL-based algorithms have comparable inference times on different datasets because they adopt the same procedure to detect anomalies (i.e., the final anomaly score is determined using MSE-based reconstruction error). Yet, the proposed PDBSNet also requires downsampling and upsampling of pixel shuffle, which takes a little extra time. Nevertheless, our PDBSNet can obtain impressive performance, which puts PDBSNet in a privileged position to be used as an HSI anomaly detector.

IV. CONCLUSION

This article presents a new PD blind-spot network for HAD. It provides new insights for adequately and reasonably utilizing limited prior information in the anomaly detection

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task. In other words, PDBSNet trains a prominent background reconstruction model in a self-supervised manner. More specifically, a blind-spot framework based on MDC is designed, i.e., for the pixel at a certain position in the output HSI of the network, the effect of the input pixel at the same position is excluded. Accordingly, this network reconstructs each pixel using the spatial–spectral characteristics of its neighbors. In addition, the PD strategy is integrated to reduce the spatial dependency between anomaly pixels, which further weakens the feature representation capability of PDBSNet for characterizing anomalies. Eventually, the anomalies in the HSI cannot be sufficiently reconstructed, leading to an increase in reconstruction errors. Naturally, the reconstruction error is applied to identify anomalies, at inference time. Experiments on four datasets demonstrate that our PDBSNet exhibits leading and comprehensive detection performance in different scenarios. However, the proposed PDBSNet may possess a limited detection effect in complex scenes containing anomalous targets and sparse background objects. Therefore, in the future, we will design new frameworks to suppress background objects with sparse distribution and improve the detection accuracy in complex scenes.

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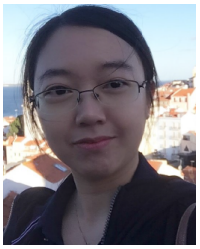

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